

Darelle Herrera

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EDUCATION

University of Houston *BS – Computer Science*

NSM Data Science Capstone (Spring - Fall 2026)

Aug 2022 – Expected Dec 2026

Coursework: **Data Structures, Algorithms, Database Systems, Software Design, Operating Systems**

Houston, TX

ACADEMIC PROJECTS

Maze Navigation Environment | Reinforcement Learning Infrastructure

Feb 2026 – May 2026

Technologies: **Python, Pillow, Numpy**

- Built a grid-based maze environment for training and evaluating reinforcement learning agents, including a graph of 4,096 navigable cells parsed from PNG maze images with dynamic hazard overlays.
- Implemented a turn-based action system with structured feedback, handling movement and hazard detection across multi-action turns.
- Engineered rotating death pit hazards that precompute 4 clockwise V-shaped configurations at load time, updating cell types incrementally each turn using set differencing for efficiency.

Skin Disease Diagnosis Using Computer Vision | AI4ALL Ignite Portfolio Project

May 2025 – Aug 2025

Technologies: **Python, CNN, ResNet, Vision Transformer(ViT), Pandas**

- Developed a deep-learning pipeline categorizing skin lesion images into 4 diagnoses, including melanoma of the skin, collaborating with 4 peers under industry mentorship.
- Achieved 80% with a supervised learning model on a database of 8 thousand images, implementing early stopping to minimize overfitting.
- Compared the efficiency of deep learning tools, including CNN, ResNet, ViT, and EfficientNet, to determine the ideal strategy, and presented results to a network of industry professionals and peers.

ACTIVITIES

Cougarettes | Association for Computing Machinery, Women Chapter (ACM-W) at UH

Jan 2026 – Present

Coding Event Coordinator

- Responsible for all technical workshops and coding-related events for an organization supporting women in STEM by mentoring students across Computer Science and Engineering disciplines.
- Recruited, trained, and managed officer interns to assist with research, presentation development, and hosting events, including LeetCode sessions and introductory programming workshops.
- Established collaborations with other organizations to expand event outreach, build cross-disciplinary networks, and nurture relationships with students to deliver relevant and engaging technical resources.

CodeRED Hackathon | CougarCS

Feb 2026 – Present

Corporate Relations and Marketing Executive Officer

- Securing sponsorships for the largest 24-hour hackathon at the University of Houston, with historical attendance of 200 - 300 people, building relationships with business executives/associates to market CodeRED and increase outreach.
- Developing marketing strategy for the event and creating a brand identity through styling guidelines to delegate tasks to team designers in creating graphics for fliers, slideshows, decorations, t-shirts, stickers, and other merchandise.
- Expanding network of attendees and professional connections through in-person meetings, virtual outreaches, and social media activity.

WORK EXPERIENCE

Coder Kids

Sept 2022 – Present

Lead Instructor

- Leading K-12 computer science and programming classes, with a focus on Scratch and Python coding, adapting to age, experience, and learning preferences of students in varying age ranges and school atmospheres.
- Planning and adapting curriculum for student needs. Responsible for tracking student progress and managing classroom equipment.
- Developed classroom-management skills for varying student ages and demographics, for year-long after-school programs, summer camps, and personalized private lessons.

TECHNOLOGIES AND CERTIFICATIONS

Programming

C++, Python, Java, SQL, R

Tools

RStudio, GitHub, VS Code, Excel, Figma, Numpy

Language

English(Bilingual), Spanish(Native Speaker)

Certifications

Microsoft Technical Resilience Program by Cougarettes at the University of Houston